

AKASH R M

AI Developer | AI, Computer Vision & Full-Stack Business Applications

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[LinkedIn](#) — [GitHub](#) — [Portfolio](#)

PROFESSIONAL SUMMARY

AI Developer with hands-on experience building AI-powered solutions, real-time computer vision systems, and full-stack business applications. Specialized in deep learning, computer vision, face recognition, and multi-object tracking, with experience developing HRMS, CRM, ERP, LMS, and attendance solutions, scalable APIs, dashboards, workflow automation, and end-to-end applications using Python, FastAPI, React, TypeScript, and PostgreSQL. Experienced in GPU-accelerated inference, real-time video processing, model optimization, and production deployment.

TECHNICAL SKILLS

- **Languages:** Python | TypeScript | JavaScript (ES2022) | SQL | HTML5 | CSS3
- **AI / Computer Vision:** Deep Learning | Machine Learning | Computer Vision | Face Detection & Recognition | Object Detection | Multi-Object Tracking (MOT) | Action Recognition | Image Classification | Model Optimization
- **ML Frameworks:** PyTorch | TensorFlow | Keras | InsightFace (SCRFD, ArcFace) | ONNX Runtime (GPU / CUDA) | OpenCV | NumPy | Pillow | YOLO
- **Backend:** FastAPI | Flask | Uvicorn (ASGI) | Pydantic | SQLAlchemy (ORM) | RESTful API Design | WebSocket APIs | JWT Auth | bcrypt
- **Frontend:** React 19 | TanStack Router | TanStack Query | React Hook Form | Zod | Tailwind CSS | Radix UI | Recharts | Vite
- **Business Applications:** HRMS | CRM | ERP | LMS | Workflow Automation | Business Process Automation | Attendance Management | Employee Management | Dashboards & Analytics
- **Video & Streaming:** RTSP | WebRTC (aiortc) | PyAV / FFmpeg | MJPEG | MSE / fMP4 | ONVIF | NVDEC / libx264 | Live Multi-Camera Pipelines
- **Databases & Cloud:** PostgreSQL | SQLite | Supabase | AWS (EC2) | Railway | Cloudflare
- **DevOps & Tools:** Docker | Git | GitHub | CI/CD

EXPERIENCE

AI Developer – GROW AI (iQue Ventures Pvt. Ltd.), Bengaluru **2025 – Present**

- Building AI-powered business solutions including HRMS, attendance management, employee analytics, and administrative dashboards.
- Developing real-time person detection, multi-object tracking, and face recognition systems for live RTSP camera streams using optimized GPU inference.
- Developing an end-to-end facial recognition attendance platform using FastAPI, PostgreSQL, React, and TypeScript, with responsive web interfaces and real-time data integration.
- Designing and implementing deep learning models and scalable ML pipelines covering data preprocessing, training, evaluation, model optimization, and deployment.
- Developing production APIs, full-stack web applications, workflow automation, and low-latency multi-camera video streaming using WebRTC and MSE/fMP4.

Assistant Technical Writer – Rpinnacle Research Solutions, Tamil Nadu **2024 – 2025**

- Produced technical documentation for deep learning and data science research projects.
- Contributed to dataset analysis, experimental evaluation, and comparative model studies.

PROJECTS

AI-Powered HRMS & Facial Recognition Attendance Platform

2026

- Developed an AI-powered workforce platform integrating facial recognition, automated attendance, payroll-related workflows, and HR/admin dashboards.
- Built the backend using FastAPI, SQLAlchemy, and PostgreSQL with JWT/bcrypt role-based authentication.
- Integrated InsightFace and ONNX Runtime with CUDA for GPU-accelerated face recognition and real-time inference.
- Implemented real-time multi-camera streaming using WebRTC, aiortc, and PyAV, with a React 19 + TanStack + Tailwind frontend and Capacitor Android application.
- Containerized the platform with Docker and deployed services using Railway and Cloudflare tunneling.

AI Real-Time Person Detection & Tracking System

2026

- Built a real-time person detection and multi-object tracking pipeline using YOLO for live RTSP camera streams with optimized inference.
- Implemented frame-wise analytics, tracking, and unique person counting for real-time monitoring.

Vision Snap – Computer Vision Dataset Tool

2025

- Built a React-based dataset-generation tool with webcam capture, video frame extraction, and automated dataset packaging workflows.

ALL IN ONE – OCR Web Application

2025

- Built a JavaScript OCR web application with modular architecture and responsive UI for structured, multi-page text and image extraction.

Hospital Management System with Brain Tumor Detection

2024

- Developed a Django-based hospital management system with authentication workflows and integrated deep-learning tumor detection and medical reporting.

INTERNSHIPS & CERTIFICATIONS

- **Python/Django Intern** – Clovion Tech Solutions Pvt. Ltd., Tamil Nadu

Jan 2024 – Mar 2024
- **Python Programming Intern** – Srishti Innovations, Technopark, Kerala

July 2023
- **In-Plant Training** – iTrobes Technologies Pvt. Ltd., Tamil Nadu

July 2022

EDUCATION

- B.E. Computer Science and Engineering**
Bethlahem Institute of Engineering, Tamil Nadu
- 2020 – 2024